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| MO-IT107: Information Management | |
|---|---|
| **Week No.** | 7 |
| **Topic:** | Logical Schema Design |
| **Worksheet No. 4** | Data Dictionary for MotorPH Payroll  System Database |

| | |
|---|---|
| **Project Manager:** | Jether Daniel A. Torres |
| **Business Analyst:** | |
| **Database Designer:** | |
| **Program and Year Level:** | BSIT - Software Development; A.Y. 2024 - 2025 |

A data dictionary assists in grasping the significance, organization, and setting of the data items by providing a clear description of the data field contained within a dataset, making it easier for everyone to work with the data.

Now that you have a better knowledge of the logical schema design and have practiced making it using the practice activity earlier, It is time to move forward with your main assignment for the week, which is also a part of the second milestone.

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**Instructions:**

1. Access your team's Milestone 1 output.
2. Using the information found in your Milestone 1, fill in the requirements needed in the sections of the table provided below.
3. Overwrite the texts marked red and delete this note prior to submission.
4. You may refer to this example as a guide.

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# *Data Dictionary for MotorPH Payroll System Database*

***Jether Daniel A. Torres***
*(Bachelor of Science in Information Technology,*
*Major in Software Development)*

## 1. Table Definitions

### 1.1. List of Database Tables

The Payroll system database is composed of the following tables:

| Group | Table | Description |
|---|---|---|
| A. Employee Details | Employee | Preserves detailed records of all employees, incorporating unique identifiers, departmental affiliations, positional data, government-linked information, and supervisory relationships |

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| Group | Table | Description |
|---|---|---|
| | Address | Maintains discrete records of employees' physical addresses, incorporating city and regional identifiers, building details, street information and postal codes |
| | Department | Delineates the various operational units within the organisation, providing unique identifiers and names that define each departmental segment |
| | Position | Registers individual job roles by associating a unique position identifier with corresponding departmental information |
| | Supervisor | Documents supervisory assignments by linking supervisor identifiers with employee records and departmental associations |
| | Schedule | Records employee scheduling data, including their start times, required daily working hours, and assigned weekly rest days, to facilitate effective workforce management and help calculate preformance in payroll |
| | Leave Balance | Tracks employee leave entitlements, maintaining balances per leave type, and recording timestamps for the latest updates to ensure accurate accrual management |
| B. Logging | Changelog | Archives logs of modifications made to employee records, including detailed timestamps, field identifiers, and both previous and updated data values for audit purposes |
| | Attendance | Registers employee attendance information, documenting clock-in and clock-out times, attendance dates and the classification of the day, thus ensuring accurate timekeeping |
| | Leave | Records employee leave requests with detailed data on leave types, durations, justifications, approval statuses and scheduled start dates to simplify absence management |
| | Leave Accrual | Records the progression of leave accruals over time by documenting accrual dates, credited leave amounts, and |

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| Group | Table | Description |
|---|---|---|
| | History | balance transitions, providing an extensive account |
| C. Government Details | Government | Consolidates government-related identifiers for each employee |
| | Government Deductibles | Itemises monthly governmental deductions, enumerating employer and employee contributions alongside withholding taxes |
| | Contributions | Records individual governmenet contributions, detailing both employee and employer shares for PhilHealth, Pag-ibig and SSS |
| D. Payroll | Payroll | Compiles extensive payroll information by linking employee identifiers with wage, rebate and summary records, while also incorporating monthly identifiers and detailed salary rates |
| | Payslip | Generates individual payslip records within the pay period, hour metrics (including overtime and undertime), and both gross and net income figures. |
| | Leave Monetisation | Documents the conversion of accrued leave into monetary value |
| | Earnings | Aggregates additional remuneration components such as overtime, paid time off, holiday pay and various stipends |
| | Deductibles | Details payroll deductions by listing rebate identifiers, other deduction totals and aggregating these values to yield the overall deductibles per payroll cycle |
| | Other Transactions | Records misc financial transactions related to payroll, such as bonuses, advances, reimbursements and additional deductions |

| Group | Table | Description |
|---|---|---|
| | Government Tally | Consolidates government contribution totals over each payroll period, providing total figures (for current payroll) of Pag-ibig, PhilHealth and SSS contributions according to designated contributor |
| | Summary | Organises an overarching financial summary of payroll disbursements, benefits, tax withholdings and government disbursements (for the month) |
| E. Static-Update Only | Regions | Maintains static records of regions by assigning a unique identifier to each region along with its designated name |
| | Cities | Lists cities by uniquely identifying each with a specific identifier, associating them with corresponding regions |
| | Holiday | Enumerates public and recognised organisational holidays by capturing unique identifiers, holiday names and corresponding dates |
| | Months | Defines monthly periods by assigning unique identifiers to each month and documenting the number of workdays |
| | Rates | Specifies statutory rates for various contributions by detailing employer and employee Pag-ibig rates, maximum contributions, PhilHealth and SSS rates |
| | Tax Table | Outlines detailed tax brackets by defining income ranges, bracket widths and associated tax rates |
| | Leave Type | Classifies different leave categories by specifying whether a leave type is mandatory, detailing default accrual rates, accrual frequencies and maximum accrual limits |

### 1.2. Column Definitions

The tables inside the MotorPH Payroll system database are composed of the following columns:

### 1.2.1. Employee Details

| A. Employee Details | | | | |
|---|---|---|---|---|
| **Employee** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| employeeID | Unique employee identifier | SmallInt | Size(5) | Primary Key, Unique |
| addressID | Linked address identifier | SmallInt | Size(5) | Foreign Key |
| departmentID | Identifier for the department | TinyInt | – | Foreign Key |
| positionID | Identifier for the position | TinyInt | – | Foreign Key |
| governmentID | Linked government record identifier | SmallInt | Size(5) | Foreign Key |
| supervisorID | Employee's immediate supervisor identifier | SmallInt | Size(5) | Foreign Key |
| firstName | Employee's given name | Varchar | Size(255) | – |
| lastName | Employee's surname | Varchar | Size(255) | – |
| contactNumber | Employee's phone number | Varchar | Size(15) | – |
| birthdate | Employee's date of birth | Date | – | CHECK (NOT NULL), |

| | | | | |
|---|---|---|---|---|
| | | | | CHECK (birthdate < CURRENT_DATE) |
| employmentStatus | Current employment status | Enum | – | – |

**Address**

| Column Name | Description | Data Type | Length and Precision | Constraints |
|---|---|---|---|---|
| addressID | Unique address record identifier | SmallInt | Size(5) | Primary Key, Unique |
| cityID | Linked city identifier | SmallInt | Size(3) | Foreign Key |
| regionID | Linked region identifier | TinyInt | Size(1) | Foreign Key |
| building | Building name or number | Varchar | Size(100) | – |
| street | Street name | Varchar | Size(150) | – |
| postalCode | Postal or zip code | Varchar | Size(5) | – |

**Department**

| Column Name | Description | Data Type | Length and Precision | Constraints |
|---|---|---|---|---|
| departmentID | Unique department identifier | TinyInt | – | Primary Key, Unique |
| departmentName | Name of the department | Varchar | Size(100) | – |

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| **Position** | | | | |
|---|---|---|---|---|
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| positionID | Unique position identifier | TinyInt | – | Primary Key, Unique |
| departmentID | Linked department identifier | TinyInt | – | Foreign Key |
| positionName | Name of the position | Varchar | Size(100) | – |
| **Supervisor** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| supervisorID | Unique supervisor record identifier | SmallInt | Size(5) | Primary Key, Unique |
| employeeID | Linked employee identifier | SmallInt | Size(5) | Foreign Key |
| departmentID | Linked department identifier | TinyInt | – | Foreign Key |
| **Schedule** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| scheduleID | Unique schedule identifier | SmallInt | Size(5) | Primary Key, Unique |
| employeeID | Linked employee identifier | SmallInt | Size(5) | Foreign Key, Unique |

| | | | | |
|---|---|---|---|---|
| startTime | Scheduled work start time | Time | – | NOT NULL |
| requiredDailyHours | Number of work hours required per day | TinyInt | Size(2) | Min(1), Max(24), NOT NULL |
| restDay | Assigned weekly rest day | SET | – | NOT NULL |
| **Leave Balance** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| balanceID | Unique leave balance identifier | Integer | – | Primary Key, Unique |
| leaveType | Linked leave type identifier | Enum | – | Foreign Key |
| scheduleID | Linked schedule identifier for employee | SmallInt | Size(5) | Foreign Key |
| currentBalance | Balance of leave type under employee | Decimal | Precision(4), Scale(2) | Min(0) |
| lastUpdated | Update timestamp | DateTime | – | – |

### 1.2.2. Logging

| **B. Logging** | | | | |
|---|---|---|---|---|
| **Changelog** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |

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| | | | | |
|---|---|---|---|---|
| changelogNo | Unique changelog entry identifier | Integer | – | Primary Key, Unique |
| employeeID | Linked employee identifier | SmallInt | Size(5) | Foreign Key |
| createdAt | Record creation timestamp | DateTime | – | NOT NULL, Default(CURRENT_TIMESTAMP) |
| appenderID | Identifier for the entry creator | SmallInt | Size(5) | – |
| updatedAt | Timestamp of the last update | DateTime | – | NOT NULL, Default(CURRENT_TIMESTAMP) |
| editorID | Identifier for the entry editor | SmallInt | Size(5) | – |
| fieldCode | Shortened identifier for Table & Column (e.g., "P_NP" for Payroll.NetPay) | Varchar | Size(10) | – |
| dataType | Defines the type of old and new values | Enum | – | – |
| oldValue | Previous value before the change, stored as text | Varchar | Size(255) | – |
| newValue | Updated value after the change, stored as text | Varchar | Size(255) | – |

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| **Attendance** | | | | |
|---|---|---|---|---|
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| attendanceID | Unique attendance record identifier | Integer | – | Primary Key, Unique |
| scheduleID | Linked schedule identifier | SmallInt | Size(5) | Foreign Key |
| date | Attendance date | Date | – | Unique (scheduleID, date) |
| timeIn | Time when the employee clocked in | Time | – | Check (timeIn <= timeOut) |
| timeOut | Time when the employee clocked out | Time | – | Check (timeIn <= timeOut) |
| dayType | Type of day | Enum | – | – |
| Leave | | | | |
| **Column Name** | Description | Data Type | Length and Precision | Constraints |
| **leaveID** | **Unique leave record identifier** | **Integer** | **–** | **Primary Key, Unique** |
| balanceID | Linked leave balance record identifier | Integer | – | Foreign Key |
| leaveType | Category of leave | Enum | – | Foreign Key |
| scheduleID | Linked schedule identifier | SmallInt | Size(5) | Foreign Key |
| dateStart | Leave start date | Date | – | NOT NULL, Check (dateStart >= CURRENT_DATE) |

| | | | | |
|---|---|---|---|---|
| hourDuration | Duration of leave in hours | SmallInt | – | Min(0) |
| leaveReason | Justification for the leave request | Varchar | Size(150) | – |
| leaveStatus | Approval status | Enum | – | – |
| requestLeaveCredit | Credits that will be consumed | Decimal | Precision(4), Scale(2) | – |
| remainingLeaveCredit | Credits remaining after used | Decimal | Precision(4), Scale(2) | – |
| dateRequested | Timestamp of leave request submission | DateTime | – | Default(CURRENT_TIMESTAMP) |
| dateApproved | NULL until request is approved | DateTime | – | NULL Allowed |
| approverID | Tracks Supervisor ID who approved leave | SmallInt | Size(5) | – |

Leave Accrual History

| **Column Name** | Description | Data Type | Length and Precision | Constraints |
|---|---|---|---|---|
| **accrualID** | **Unique leave accrual record identifier** | **Integer** | **–** | **Primary Key, Unique** |
| balanceID | Linked leave balance record identifier | Integer | – | Foreign Key |
| leaveType | Category of leave for accrual purposes | Enum | – | NOT NULL |
| accrualDate | Date when the leave accrual is | Date | – | NOT NULL, Check |

| | | | | |
|---|---|---|---|---|
| | recorded and applied | | | (accrualDate ≤ CURRENT_DATE) |
| creditAccrued | Number of leave credits accrued until accrual date | Decimal | Precision(4), Scale(2) | Min(0) |
| oldBalance | Leave balance prior to the accrual | Decimal | Precision(4), Scale(2) | Min(0) |
| newBalance | Leave balance subsequent to the accrual | Decimal | Precision(4), Scale(2) | Min(0) |

### 1.2.3. Government Details

| C. Government Details | | | | |
|---|---|---|---|---|
| **Government** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| governmentID | Unique government record identifier | SmallInt | Size(5) | Primary Key, Unique |
| employeeID | Linked employee identifier | SmallInt | Size(5) | Foreign Key |
| TIN | Taxpayer Identification Number | Varchar | Size(15) | – |
| sssNo | Social Security System number | Varchar | Size(12) | – |
| philHealthNo | PhilHealth identifier | Varchar | Size(12) | – |

| | | | | |
|---|---|---|---|---|
| pagIbigNo | Pag-ibig fund identifier | Varchar | Size(12) | – |
| encryptedRef | Encrypted reference data | Varbinary | Max | – |
| **Government Deductibles** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| governmentID | Linked government record identifier | SmallInt | Size(5) | Primary Key, Foreign Key |
| monthID | Linked month identifier | Enum | – | Foreign Key |
| employerContribution | Employer contribution amount | Decimal | Precision(19), Scale(4) | Min(0) |
| employeeContribution | Employee contribution amount | Decimal | Precision(19), Scale(4) | Min(0) |
| withholdingTax | Amount of tax withheld | Decimal | Precision(19), Scale(4) | Min(0) |
| **Contributions** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| governmentID | Linked government record identifier | SmallInt | Size(5) | Composite Primary Key, Foreign Key |
| contributor | Identifier for contributor | Enum | – | Composite Primary Key |
| philHealthContributi | PhilHealth contribution amount | Decimal | Precision(19), Scale(4) | Min(0) |

| | | | | |
|---|---|---|---|---|
| on | | | | |
| pagIbigContribution | Pag-ibig contribution amount | Decimal | Precision(19), Scale(4) | Min(0) |
| sssContribution | SSS contribution amount | Decimal | Precision(19), Scale(4) | Min(0) |

### 1.2.4. Payroll

| D. Payroll | | | | |
|---|---|---|---|---|
| **Payroll** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| payrollID | Unique payroll record identifier | Integer | – | Primary Key, Unique |
| employeeID | Linked employee identifier | SmallInt | Size(5) | Foreign Key |
| rebateID | Linked rebate record identifier | Integer | – | Foreign Key |
| wageID | Linked wage record identifier | Integer | – | Foreign Key |
| summaryID | Linked summary record identifier | Integer | – | Foreign Key |
| monthID | Identifier for the payroll month | Enum | – | – |
| ptoMax | Maximum allowed paid time off (PTO) | Integer | – | Min(0) |
| hourlyRate | Pay rate per hour | Decimal | Precision(19), Scale(4) | Min(0), NOT NULL |

| | | | | |
|---|---|---|---|---|
| dailyRate | Fixed daily pay rate | Decimal | Precision(19), Scale(4) | Min(0), NOT NULL |
| monthlyRate | Monthly salary rate | Decimal | Precision(19), Scale(4) | Min(0) |
| netPay | Final pay after deductions | Decimal | Precision(19), Scale(4) | Min(0), NOT NULL |

**Payslip**

| Column Name | Description | Data Type | Length and Precision | Constraints |
|---|---|---|---|---|
| payslipID | Unique payslip identifier | Integer | – | Primary Key, Unique |
| payrollID | Linked payroll record identifier | Integer | – | Foreign Key |
| periodStartDate | Start date of the pay period | Date | – | NOT NULL |
| periodEndDate | End date of the pay period | Date | – | Unique (payrollID, periodEndDate) |
| regularHours | Standard work hours | SmallInt | – | Min(0), NOT NULL |
| overtimeHours | Overtime hours worked | SmallInt | – | Min(0) |
| holidayOvertimeHours | Overtime during holidays | SmallInt | – | Min(0) |
| ptoHours | Total paid time off hours taken | SmallInt | – | Min(0) |
| undertimeHours | Hours below expected work time | SmallInt | – | Min(0) |
| daysWorked | Total days worked | TinyInt | – | Min(0) |

| | | | | |
|---|---|---|---|---|
| holidaysWorked | Number of holidays worked | TinyInt | – | Min(0) |
| restDaysWorked | Number of rest days worked | TinyInt | – | Min(0) |
| grossIncome | Total earnings before deductions | Decimal | Precision(19), Scale(4) | Min(0) |

**Leave Monetisation**

| Column Name | Description | Data Type | Length and Precision | Constraints |
|---|---|---|---|---|
| monetisationID | Unique leave monetisation record identifier | Integer | – | Primary Key, Unique |
| leaveType | Type of leave being monetised | Enum | – | NOT NULL |
| balanceID | Linked leave balance record identifier for monetisation | Integer | – | Foreign Key |
| creditsMonetised | Number of leave credits converted to monetary value | Decimal | Precision(4), Scale(2) | Min(0) |
| monetaryValuePer | Monetary value assigned per leave credit | Decimal | Precision(19), Scale(4) | Min(0) |
| totalConverted | Aggregate monetary value resulting from the conversion | Decimal | Precision(19), Scale(4) | Min(0) |
| dateMonetised | Date when leave monetisation was effected | Date | – | – |

| | | | | |
|---|---|---|---|---|
| approverID | Identifier for the approver responsible for validating the monetisation | SmallInt | Size(5) | – |
| **Earnings** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| wageID | Unique wage record identifier | Integer | – | Primary Key, Unique |
| overtimePay | Payment for overtime hours | Decimal | Precision(19), Scale(4) | Min(0) |
| ptoPay | Paid time off compensation | Decimal | Precision(19), Scale(4) | Min(0) |
| holidayPay | Payment for holiday work | Decimal | Precision(19), Scale(4) | Min(0) |
| restDayPay | Payment for work on rest days | Decimal | Precision(19), Scale(4) | Min(0) |
| riceSubsidy | Subsidy provided for rice | Decimal | Precision(19), Scale(4) | Min(0) |
| phoneStipend | Stipend for phone expenses | Decimal | Precision(19), Scale(4) | Min(0) |
| clothesStipend | Stipend for clothing expenses | Decimal | Precision(19), Scale(4) | Min(0) |
| otherIncentTotal | Sum of other incentives | Decimal | Precision(19), Scale(4) | Min(0) |
| totalBenefits | Total benefit amount | Decimal | Precision(19), Scale(4) | Min(0) |
| **Deductibles** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| rebateID | Unique rebate record identifier | Integer | – | Primary Key, Unique |

| | | | | |
|---|---|---|---|---|
| governmentID | Linked government record identifier | SmallInt | Size(5) | Foreign Key |
| otherDeductTotal | Sum of other deductions | Decimal | Precision(19), Scale(4) | Min(0) |
| totalDeductibles | Overall deductions sum | Decimal | Precision(19), Scale(4) | Min(0) |
| **Other Transactions** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| transactionID | Unique transaction record identifier | Integer | – | Primary Key, Unique |
| payrollID | Payroll identifier for the transaction | Integer | – | Foreign Key |
| transactionType | Type of transaction | Enum | – | – |
| transactionAmount | Amount of the transaction | Decimal | Precision(19), Scale(4) | Min(0) |
| description | Description of the transaction | Varchar | Size(150) | – |
| **Government Tally** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| summaryID | Linked summary record identifier | Integer | – | Primary Key (assumed) |
| contributor | Identifier for contributor | Enum | – | – |
| overallPagIbig | Overall (between this month's payrolls) Pag-ibig contribution | Decimal | Precision(19), Scale(4) | Min(0) |

| | | | | |
|---|---|---|---|---|
| overallPhilHealth | Overall (between this month's payrolls) PhilHealth contribution | Decimal | Precision(19), Scale(4) | Min(0) |
| overallSSS | Overall (between this month's payrolls) SSS contribution | Decimal | Precision(19), Scale(4) | Min(0) |
| **Summary** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| summaryID | Unique summary record identifier | Integer | – | Primary Key, Unique |
| monthID | Linked month identifier | Enum | – | – |
| overallGrossIncome | Total gross income summed | Decimal | Precision(19), Scale(4) | Min(0) |
| overallBenefits | Total benefits summed | Decimal | Precision(19), Scale(4) | Min(0) |
| employeeDisbursementTotal | Total disbursements to employees | Decimal | Precision(19), Scale(4) | Min(0) |
| employerDisbursementTotal | Total amount disbursed by the employer | Decimal | Precision(19), Scale(4) | Min(0) |
| overallPITotal | Combined Pag-ibig total | Decimal | Precision(19), Scale(4) | Min(0) |
| overallPHTotal | Combined PhilHealth total | Decimal | Precision(19), Scale(4) | Min(0) |
| overallSSSTotal | Combined SSS total | Decimal | Precision(19), Scale(4) | Min(0) |

| | | | | |
|---|---|---|---|---|
| overallTaxTotal | Total tax withheld summed | Decimal | Precision(19), Scale(4) | Min(0) |
| governmentDisbursementTotal | Total government disbursement summed | Decimal | Precision(19), Scale(4) | Min(0) |

### 1.2.5. Static-Update Only

| F. Static-Update Only | | | | |
|---|---|---|---|---|
| **Regions** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| regionID | Unique region identifier | TinyInt | Size(1) | Primary Key, Unique |
| regionName | Name of the region | Varchar | Size(100) | – |
| **Cities** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| cityID | Unique city identifier | SmallInt | Size(3) | Primary Key, Unique |
| regionID | Linked region identifier | TinyInt | – | Foreign Key |
| cityName | Name of the city | Varchar | Size(100) | Unique |

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| Holiday | | | | |
|---|---|---|---|---|
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| holidayID | Unique holiday record identifier | TinyInt | – | Primary Key, Unique |
| holidayName | Name of the holiday | Varchar | Size(255) | Unique |
| holidayDate | Date of the holiday | Date | – | Unique |
| **Months** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| monthID | Unique month identifier | Enum | – | – |
| monthlyWorkdays | Number of workdays in the month | Integer | Size(2) | Min(1), Max(31) |
| **Rates** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| piEmployerRate | Global employer Pag-ibig rate | Decimal | Precision(5), Scale(3) | Min(0), NOT NULL |
| piEmployeeRate | Global employee Pag-ibig rate | Decimal | Precision(5), Scale(3) | Min(0), NOT NULL |
| piMaxContribution | Maximum Pag-ibig contribution | Decimal | Precision(5), Scale(3) | Min(0), NOT NULL |
| phRate | Global PhilHealth rate | Decimal | Precision(5), Scale(3) | Min(0), NOT NULL |
| sssRate | Global SSS rate | Decimal | Precision(5), Scale(3) | Min(0), NOT NULL |

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| Tax Table | | | | |
|---|---|---|---|---|
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| taxBracket | Identifier for the tax bracket | Integer | Min(1) | Primary Key |
| minCompensation | Minimum income for the bracket | Decimal | Precision(19), Scale(4) | Min(0), NOT NULL |
| maxCompensation | Maximum income for the bracket | Decimal | Precision(19), Scale(4) | Min(0), NOT NULL |
| bracketWidth | Income range width for the bracket | Decimal | Precision(19), Scale(4) | Min(0) |
| taxRate | Tax rate for the bracket | Decimal | Precision(19), Scale(4) | Min(0), NOT NULL |
| **Leave Type** | | | | |
| **Column Name** | **Description** | **Data Type** | **Length and Precision** | **Constraints** |
| leaveType | Classification of leave | Enum | – | NOT NULL |
| mandatory | Indicates whether the leave type is mandatory | Boolean | – | – |
| defaultAccrualRate | Standard rate at which leave accrues for this type | Decimal | Precision(4), Scale(2) | Min(0) |
| accrualFrequency | Frequency of leave accrual | Enum | – | – |
| maxAccrual | Maximum allowable accrual for the leave type | Decimal | Precision(4), Scale(2) | Min(0) |

## 2. Relationships

| Foreign Key Table | Foreign Key Column | Referenced Table | Referenced Primary Key | Description of Relationship |
|---|---|---|---|---|
| Employee | addressID | Address | addressID | Each employee is assigned an address (1:1) |
| | departmentID | Department | departmentID | Employees are categorised by their department (N:1) |
| | positionID | Position | positionID | The employee holds a specific position (N:1) |
| | governmentID | Government | governmentID | Government records are uniquely linked to each employee (1:1) |
| | supervisorID | Supervisor | supervisorID | The system references the supervisor of the employee (N:1) |
| Address | cityID | Cities | cityID | Each address entry is located within a specific city (1:1) |
| | regionID | Regions | regionID | Addresses are also linked to a region (1:1) |
| Position | departmentID | Department | departmentID | Positions are assigned within departments (N:1) |
| Supervisor | | | | Departments have at least one supervisor (N:1) |
| | employeeID | Employee | employeeID | Supervisors are employees themselves, connecting both tables directly (1:1) |
| Schedule | | | | Each employee is assigned a schedule (1:1) |
| Changelog | | | | Tracks changes made by the employee (N:1) |

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| Foreign Key Table | Foreign Key Column | Referenced Table | Referenced Primary Key | Description of Relationship |
|---|---|---|---|---|
| LeaveBalance | leaveType | Leave Type | leaveType | Indicates the leave type applicable for employee leave balance (N:1) |
| LeaveBalance | scheduleID | Schedule | scheduleID | Leave balances are determined based on an employee's schedule (N:1) |
| Attendance | scheduleID | Schedule | scheduleID | Attendance records are linked to the schedule (N:1) |
| Leave | scheduleID | Schedule | scheduleID | Leave requester identifier (N:1) |
| Leave | leaveType | Leave Type | leaveType | Indicates the leave type used for request (N:1) |
| Leave | balanceID | Leave Balance | balanceID | Leave requests are validated against available leave balances (N:1) |
| Leave Accrual History | balanceID | Leave Balance | balanceID | Accrual records adjust the leave balance over time (N:1) |
| Leave Accrual History | leaveType | Leave Type | leaveType | Indicates the leave type has been accrued (N:1) |
| Leave Monetisation | leaveType | Leave Type | leaveType | Indicates the leave type is requested for conversion (N:1) |
| Leave Monetisation | balanceID | Leave Balance | balanceID | Converts leave credits into monetary value based on the leave balance (N:1) |
| Government | employeeID | Employee | employeeID | Links each government record with an employee (1:1) |
| Payroll | employeeID | Employee | employeeID | Links payroll records to employees (N:1) |
| Payroll | rebateID | Deductibles | rebateID | Deduction computations of payroll (1:1) |

| Foreign Key Table | Foreign Key Column | Referenced Table | Referenced Primary Key | Description of Relationship |
|---|---|---|---|---|
| | wageID | Earnings | wageID | Benefit computations of payroll (1:1) |
| | summaryID | Summary | summaryID | Collates payroll data into a monthly summary record (N:1) |
| | monthID | Months | monthID | References payroll month and its corresponding working days (1:1) |
| Government Deductibles | | | | Reference the month these deductibles are for (N:1) |
| | governmentID | Government | governmentID | Associates deductibles with a government record (1:1) |
| Contributions | | Government Deductibles | | Reference location where computed contribution will be collated (N:1) |
| Deductibles | | Government Deductibles | | Needs government deductibles to compute total deductibles (1:1) |
| Payslip | payrollID | Payroll | payrollID | Each payslip is uniquely tied to a payroll record (1:1) |
| Other Transactions | | | | Connects misc transactions to payroll records (N:1). |
| Government Tally | summaryID | Summary | summaryID | Aggregates government-related disbursements within a summary (N:1). |

## 3. Definition of Terms

Listed below are the key terms and acronyms used throughout this document.

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

| Term | Definition |
|---|---|
| TIN | Taxpayer Identification Number – a unique number assigned to individuals or entities for tax purposes. |
| SSS | Social Security System Number – an identifier used to track social security contributions and benefits. |
| PhilHealth | Identifier for the Philippine Health Insurance Corporation – used to manage health insurance records and contributions. |
| PagIbig | Identifier for the Pag-IBIG Fund (Home Development Mutual Fund) – used for housing savings and related benefits. |
| PTO | Paid Time Off – the policy allowing employees to take leave while still receiving their regular pay. |
| PI | No, not Archimedes' constant. Abbreviation used in rate fields (piEmployerRate, piEmployeeRate, piMaxContribution) – refers to Pag-IBIG related contribution rates. |
| fieldCode | A shortened identifier used in logging to represent specific table and column names concisely (e.g., "P_NP" for Payroll.NetPay). |
| Varchar | Variable Character – a data type for storing text strings of varying lengths. |
| Varbinary | Variable Binary – a data type for storing binary data that does not have a fixed length. |
| Enum | Enumeration – a data type that defines a set of predefined values. |
| TinyInt | Tiny Integer – a data type representing a small-range integer (values that typically won't |

| Term | Definition |
|---|---|
| | reach beyond 127) |
| SmallInt | Small Integer – a numeric data type used to store integers within a limited range (values that typically won't reach beyond 32,767) |
| Precision | The total number of digits allowed in a decimal number, both to the left and right of the decimal point |
| Scale | The number of digits allowed to the right of the decimal point in a decimal number. |
| accrual | The process by which leave credits are accumulated over time, typically based on a defined rate or frequency. |
| monetisation | The conversion of leave credits into monetary value, allowing employees to receive compensation for unused leave. |
| bracketWidth | Represents the income range covered by a tax bracket, defining the span between the minimum and maximum compensation for that bracket. |
| contributor | Identifier for the party (such as an employee or employer) contributing to a benefit or deduction in payroll and related calculations. |